Lightweight Formation Control Algorithm for Fix-Shaped Formations to Avoid Collision

***Abstract*— This documentation serves as a basic description of the proposed algorithm for centralised formation control, where the formation is allowed to scale and rotate to avoid obstacles.**

***Index Terms*— formation control, mixed-integer programming, centralised optimization**

I. Problem formulation

In this work, the goal is to guide a formation of vehicles along a pre-generated trajectory $r_\mathrm{f}(t) \in \mathbb{R}^2;\ \ t \in [0, T]$ and avoid collision with obstacles along the path. The formation is defined by a star-shaped polygon and its vertices, denoted by $V_v \in \mathbb{R}^2;\ \ v \in \mathbb{I}_0^{N_\mathrm{v}}$, where N_v denotes the number of vertices and $\mathbb{I}_{\ell_1}^{\ell_2}$ is being defined as the index set $\{\ell \in \mathbb{Z} \mid \ell_1 \leq \ell \leq \ell_2\}$. The polygon's vertices, along with the waypoints and the position of the obstacles are all defined in the Frenet frame, centered at $r_\mathrm{f}(t)$.

The trajectories of the vehicles are defined by B-splines. The position trajectory of vehicle i is represented by $\hat{x}_i(t) \in \mathbb{R}^2;\quad i \in \mathbb{I}_0^{N_\mathrm{i}-1}$, where N_i denotes the number of vehicles. The neighbour-set of vehicle i is denoted by $\mathcal{N}_i$. Constraints are fulfilled at N_r linearly spaced time instances: $\bar{t} = [t_0\ t_1\ \ldots t_{N_\mathrm{r}-1}]$. The symbol $t \in [t_0, t_{N_\mathrm{r}-1}]$ denotes time. The reference vector pointing from vehicle j to vehicle i is represented by $x_{i,j}^\mathrm{ref} \in \mathbb{R}^2$, defined as $x_{i,j}^\mathrm{ref} = V_i - V_j$. The formation is allowed to scale and rotate. The vehicles are assumed to be guided by double integrator dynamics.

II. Proposed method

Ordinary obstacle avoidance has been shown to work well using mixed integer programming. Formation control however raises interesting questions on how to formulate the optimization problem to allow rotation and scaling of the formation while avoiding obstacles and still being performant. In this section we will show how to formulate these constraints, starting with the proposed formulation for rotation and scaling in Section II-A, followed by Section II-B, presenting the accompanying formation constraint. In Section II-C we present the necessary equations for formulating a collision avoidance constraint in a moving Frenet frame, then in Section II-D the cost used in the algorithm. Finally, Section II-E summarizes the full optimization model.

A. Rotation and scaling of the formation

Rotation of the formation may be described by a spline variable $\hat{\phi}(t)$. However, using it directly and expressing rotation via a rotation matrix in the optimization is not possible using some solvers, like Gurobi, where trigonometric expressions are not allowed.

A possible solution for a mixed integer formulation of the problem would be using binary variables to choose between predefined angles (whose sine and cosine values are pre-calculated). To control the scaling or shrinking of the formation, additional binary variables or a continuous variable can be defined separately. The number of these binary variables however drastically increases when finer rotation of the formation is desired, leading to a significant performance hit. For this reason, a new method for these kinds of problem is proposed in the following.

We propose to describe the rotation and scaling of the formation with two variables, intended to encode scaled cosine and sine values. These two values are: $c \in \mathbb{R}_+^{N_\mathrm{r}}, s \in \mathbb{R}_+^{N_\mathrm{r}}$, with $c = [c_0\ c_1\ \ldots c_{N_\mathrm{r}-1}]^\top$ and $s = [s_0\ s_1\ \ldots s_{N_\mathrm{r}-1}]^\top$ respectively.

This formulation will allow rotation in the range $[0, \frac{\pi}{2}]$, but the method can be extended to all four quadrants using binary variables without significant change to the algorithm.

These variables will form a matrix $R^\star : \mathbb{R}^2 \mapsto \mathbb{R}^{2\times 2}$ at time index $\tau \in \mathbb{I}_0^{N_\mathrm{r}-1}$:

$$R^\star(c_\tau, s_\tau) = \begin{bmatrix} c_\tau & -s_\tau \\ s_\tau & c_\tau \end{bmatrix}. \tag{1}$$

$R^\star(\cdot,\cdot)$ resembles a rotation matrix, however because its determinant is not equal to 1, it is not a true rotation matrix. In fact, it also incorporates a scaling of $\det(R^\star)^{\frac{1}{2}}$.

To limit the scaling of $R^\star$, additional constraints have to be defined. In a real-world problem, the allowed scaling of the formation is usually lower bound to avoid inter-agent collision and upper bound for example due to the length of the cable in a collaborative transportation problem. To limit scaling, a constraint like (2) can be implemented. This allows the reduction of size by a factor of $\theta_\mathrm{min} \in \mathbb{R}_+$ and a maximal increase in size by a factor of $\theta_\mathrm{max} \in \mathbb{R}_+$.

$$\theta_\mathrm{min}^2 \leq \det(R^\star) \leq \theta_\mathrm{max}^2. \tag{2}$$

Constraint (2) is however nonlinear and difficult to solve. A relaxation of it is however possible.

Consider Figure 1. Constraint (2) limits the solution space between the two quarter circles, which limit the maximal contraction to θ_min times the original size and the maximal expansion to θ_max times the original size. Constraint (2) can be relaxed using a linear approximation of it, as shown in Figure 1. As a safety measure, contraction should never be lower than the preset value (so as to avoid potential collision between vehicles). Therefore the lower bound is chosen such, that it will always be fulfilled. This is illustrated by the line

named $y_{\min}$ in the Figure. Similarly, is the upper constraint enforced. Here we approximate the upper bound by $y_{\max}$, shown in Figure 1. This relaxation is also conservative, but it ensures, that the scaling never exceeds the maximum allowed amount. This is important, because if the drones are carrying a payload, their distance is hard-constrained by the length of the ropes connecting them. The relaxed constraint is shown in (3):

$$\frac{2}{\sqrt{2}}\theta_{\min} \leq c_\tau + s_\tau \leq \theta_{\max}. \tag{3}$$

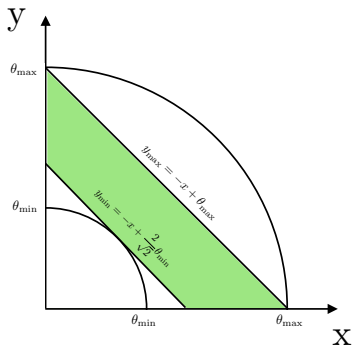


Fig. 1: Constraint relaxation

B. Formation constraint

Maintaining strict formation is desired in many cases, especially in collaborative transportation tasks, where the balance of the transported load also plays an important role. Using the variables introduced in Section II-A a simple constraint can be used to enable strict formation keeping, while allowing the group to scale and rotate. The linear constraint defined in (4) fulfills both desires.

$$\begin{aligned} g_i^{\text{form}}(\hat{x}_i(t_\tau),&\hat{x}_j(t_\tau), c_\tau, s_\tau) = \\ &(\hat{x}_i(t_\tau) - \hat{x}_j(t_\tau)) - R(c_\tau, s_\tau) \cdot x_{i,j}^{\text{ref}} = 0. \end{aligned} \tag{4}$$

In (4) the vector pointing from vehicle i to vehicle j is restricted to be parallel (due to the vector product) to the rotated-scaled reference vector of the same two vehicles. Thus, the formation is able to rotate, scale, but still able to retain its original shape.

C. Collision avoidance

The collision avoidance constraints are based on [1]. The method poses four inequality constraints (2 for each axes) with regards to the arrangement of the vehicle and the particular obstacle k, with $k \in \mathbb{I}_0^{N_\text{k}-1}$, where N_k denotes the number of obstacles. Let us denote the x coordinate of the ith vehicle at time index τ by $x_{i,\tau,\text{x}} \in \mathbb{R}$ and with $x_{i,\tau,\text{y}} \in \mathbb{R}$ its y coordinate. Furthermore, let us denote the x and y coordinates of the kth obstacle at time index τ with $q_{k,\tau,\text{x}} \in \mathbb{R}$ and $q_{k,\tau,\text{y}} \in \mathbb{R}$ respectively, and its width and height with $d_{k,\text{x}} \in \mathbb{R}_+$ and $d_{k,\text{y}} \in \mathbb{R}_+$ respectively.

To apply the method to the current problem, a slight modification to the equations presented in [1] are required. In [1] the obstacles are stationary and their sides are aligned with the coordinate axes. In our problem however, due to the moving and rotating Frenet frame, even stationary obstacles are seen as moving and rotating obstacles. Let us denote the tilting of obstacle k at index τ by the angle value $\beta_{k,\tau}$. Consequently, by rotating the vehicle and the obstacle by $-\beta_{k,\tau}$ using the rotation matrix $\Gamma \in SO(2)$ defined in (5), the collision-avoidance constraints can be constructed. Let us denote the x coordinate of vehicle i at τ, when corrected for the tilt of obstacle k with $x'_{i,\tau,k,\text{x}}$ and its y coordinate with $x'_{i,\tau,k,\text{y}}$, defined in (6a) and (6b) respectively. Similarly do we define the rotated position values of obstacle k by (6b) and (6c) respectively.

$$\Gamma(\beta) = \begin{bmatrix} \cos(\beta) & -\sin(\beta) \\ \sin(\beta) & \cos(\beta) \end{bmatrix}. \tag{5}$$

$$x'_{i,\tau,k,\text{x}} = \cos(-\beta_{k,\tau})x_{i,\tau,\text{x}} - \sin(-\beta_{k,\tau})x_{i,\tau,\text{y}}, \tag{6a}$$
$$x'_{i,\tau,k,\text{y}} = \sin(-\beta_{k,\tau})x_{i,\tau,\text{x}} + \cos(-\beta_{k,\tau})x_{i,\tau,\text{y}}, \tag{6b}$$
$$q'_{k,\tau,\text{x}} = \cos(-\beta_{k,\tau})q_{k,\tau,\text{x}} - \sin(-\beta_{k,\tau})q_{k,\tau,\text{y}}, \tag{6c}$$
$$q'_{k,\tau,\text{y}} = \sin(-\beta_{k,\tau})q_{k,\tau,\text{x}} + \cos(-\beta_{k,\tau})q_{k,\tau,\text{y}}, \tag{6d}$$

Consequently, the collision avoidance constraints are defined as:

$$\begin{aligned} x'_{i,\tau,k,\text{x}} - (q'_{k,\tau,\text{x}} + d_{k,\text{x}}) &\geq d^\star - Mc_{i,k,\tau,1}, \\ -x'_{i,\tau,k,\text{x}} + (q'_{k,\tau,\text{x}} + d_{k,\text{x}}) &\geq d^\star - Mc_{i,k,\tau,0}, \\ x'_{i,\tau,k,\text{y}} - (q'_{k,\tau,\text{y}} + d_{k,\text{y}}) &\geq d^\star - Mc_{i,k,\tau,3}, \\ -x'_{i,\tau,k,\text{y}} + (q'_{k,\tau,\text{y}} + d_{k,\text{y}}) &\geq d^\star - Mc_{i,k,\tau,2}, \\ \sum_{\ell=0}^{3} c_{i,k,\tau,\ell} &\leq 3, \\ c_{i,k,\tau,\ell} &\in \{0,1\}, \\ \forall i \in \mathbb{I}_0^{N_i-1}; \forall k \in \mathbb{I}_0^{N_k-1}; &\forall \tau \in \mathbb{I}_0^{N_\tau-1}; \forall \ell \in \mathbb{I}_0^3, \end{aligned} \tag{7}$$

where $d^\star \in \mathbb{R}_+$ is the safety distance, $M \in \mathbb{R}_+$ is a large value, larger than any position value encountered in the problem and $c_{i,k,\tau,\ell} \in \{0,1\}$ are binary variables. The value of $c_{i,k,\tau,\ell}$ determines which of the equations can be relaxed. However, in order for the collision avoidance to be met for indices i, k, τ, one ℓ of $c_{\tau,i,k,\ell}$ has to be zero. This indicates, that there is at least $d^\star$ distance between the obstacle and the vehicle.

D. Cost function

TO find the optimal trajectories, a cost function is defined, which punishes the acceleration of the vehicles w.r.t. the Frenet frame:

$$J_i^{\text{acc}}(\ddot{\hat{x}}_i(t)) = \int_0^{T_h} \alpha_\text{a} \|\ddot{\hat{x}}_i(t)\|_2^2 \, dt, \tag{8}$$

where $\alpha_\text{a} \in \mathbb{R}$ is the acceleration penalty cost.

Furthermore, large changes in rotation and (because scaling is incorporated in c_τ and s_τ) large changes is scaling can be avoided using:

$$J_{\text{rot}}(c_\tau, s_\tau) = \alpha_{\text{scale}} \sum_{\tau=1}^{N_\text{r}-1} (c_\tau - c_{\tau-1})^2 + (s_\tau - s_{\tau-1})^2. \tag{9}$$

However a preference for scaling or rotation cannot be arbitrarily set, as these are coupled together in c_τ and s_τ.

E. Optimization model

Thus, the centralised mixed-integer program becomes:

$$\begin{aligned} \underset{\vartheta}{\text{minimise}} \quad & J_{\text{rot}}(c_\tau, s_\tau) + \frac{1}{N_\text{i}} \sum_{i=0}^{N_\text{i}-1} J_i^{\text{acc}}(\ddot{\hat{x}}_i(t)) \\ \text{subject to} \quad & x_i(t_\tau) \in \mathcal{X}_{\text{free}} \\ & \frac{2}{\sqrt{2}}\,\theta_{\min} \le c_\tau + s_\tau \le \theta_{\max} \\ & g_i^{\text{form}}(\hat{x}_i(t_\tau), \hat{x}_j(t_\tau), c_\tau, s_\tau) = 0 \\ & \forall i \in \mathbb{I}_0^{N_\text{i}-1};\ \ \forall j \in \mathcal{N}_i;\ \ \forall \tau \in \mathbb{I}_0^{N_\text{r}-1}, \end{aligned} \tag{10}$$

where $x_i(t_\tau) \in \mathcal{X}_{\text{free}}$ means the collision avoidance constraint defined in (7) and ϑ collects all decision variables (spline coefficients of all the vehicle trajectories, c and s variables) to a single vector.

III. Results

The problem defined in (10) is a profoundly expressive solution to the formation control problem. Two scenarios were compared. First, Figure 2 shows results for the case, when the determinant of the $R^\star(\cdot,\cdot)$ defined in (1) is constrained. Second, Figure 3 shows the relaxed formulation with the linear constraint. Calculation times and overall cost are summarized in Table I. It can be concluded, that the relaxed version is two magnitudes faster and is capable of creating trajectories with near identical cost to the non-relaxed version.

TABLE I: Calculation times

version type	constraint	calc. time (s)	final cost
non-relaxed	$\theta_{\min}^2 \le \det(R^\star) \le \theta_{\max}^2$	22	4.3153
relaxed	$\frac{\sqrt{2}}{2}\theta_{\min} \le c_\tau + s_\tau \le \theta_{\max}$	0.12	4.85781

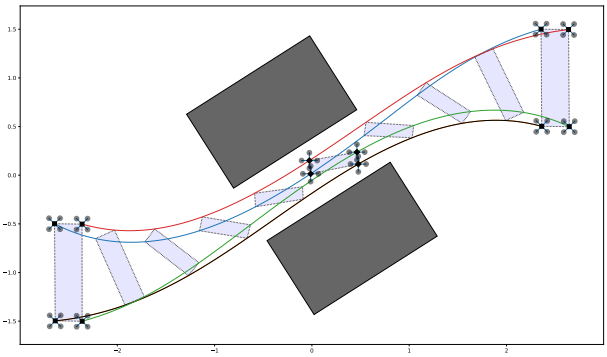


Fig. 2: Constraint: $\theta_{\min}^2 \le \det(R^\star) \le \theta_{\max}^2$. Total calculation time: 22 seconds. Final cost: 4.3153

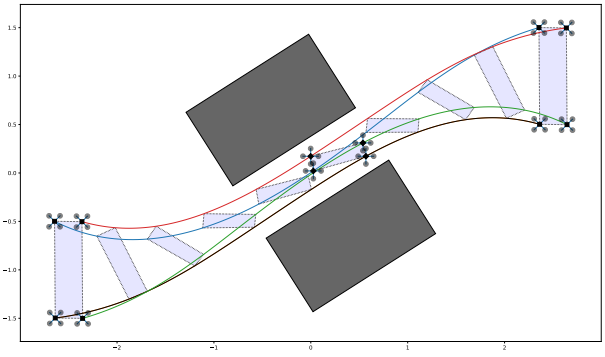


Fig. 3: Using relaxed constraint: $\frac{\sqrt{2}}{2}\theta_{\min} \le c_\tau + s_\tau \le \theta_{\max}$. Total calculation time: 0.12 seconds. Final cost: 4.85781

References

[1] A. Richards and J. P. How, "Aircraft trajectory planning with collision avoidance using mixed integer linear programming," in *Proceedings of the 2002 American Control Conference (IEEE Cat. No.CH37301)*, vol. 3, pp. 1936–1941 vol.3, 2002.